

Dataset of Chingola Municipal Council Information Extracted from Its Official Website

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Abstract

This data article describes a dataset collected from the official website of Chingola Municipal Council in Zambia. The dataset was developed as part of the CSC 4792 Data Mining and Warehousing mini-project (Team 7) and contains structured information extracted from the council's publicly available digital resources. The data collection focused on information relevant to local authority functions, including council projects, Constituency Development Fund (CDF) applications and projects, budgets, revenue efficiency, internal audits, capital expenditure, and integrated development planning. The extracted data were cleaned and processed using Python in Google Colab, transforming 23 raw files into structured comma-separated value (CSV) files using the pipe character (|) as the field separator. Data preprocessing included standardizing snake_case column names, removing missing value placeholders, currency cleaning (converting ZMW strings to numeric values), deduplication, and quality assessment across 185 total features. The resulting dataset provides structured local-government data for research, visualization, and data mining applications.

Specifications Table

Item	Description
Subject area	Data Mining and Warehousing
Specific subject area	Local government and municipal council data mining
Geographical coverage	Chingola, Copperbelt Province, Zambia
Data collection period	September 2026

Item	Description
Type of data	Tables, structured text, numerical data, categorical data, dates
Data acquisition	Automated and manual web extraction from Chingola Municipal Council digital documents and pages
Data source location	Chingola Municipal Council Official Website (https://www.chingolacouncil.gov.zm)
Data accessibility	Publicly accessible via Kaggle dataset repository and GitHub codebase
Data format	CSV files using the pipe character () as field delimiter
Data processing	Column standardization (snake_case), string currency sanitization (ZMW), missing value handling, duplicate removal, and numeric type conversion
Related project	CSC 4792 Data Mining and Warehousing Mini Project (TEAM 7)

Value of the Data

- **Structured Municipal Governance Data:** Converts unstructured council publications (audits, CDF lists, budgets) into 23 pipe-delimited CSV tables ready for analytical pipelines.
- **Financial & Project Analytics:** Enables exploratory data analysis into council revenue efficiency, internal audit compliance, CDF project distribution, and budget execution across financial years.
- **Data Quality Pipeline Reference:** Offers a reusable Python/Pandas preprocessing script for standardizing messy public-sector tabular data (snake_case conversion, regex string cleaning, missing value mapping).
- **Comparative Municipal Research:** Serves as a baseline dataset for academic benchmarking against other local authorities in Zambia.

Data Description

The dataset contains structured municipal records extracted from official documents and pages hosted on the Chingola Municipal Council portal. The final collection comprises 23 CSV files covering 185 total column attributes.

Dataset Summary Inventory

Dataset Filename	Rows	Columns
db-unza26-csc4792-chingola_citizen_engagement_strategy_objectives.csv	4	4
db-unza26-csc4792-chingola_idp_national_support_financing_2019_2023.csv	5	18
db-unza26-csc4792-chingola_idp_selected_capital_investment_plan_2024_2033.csv	5	5
db-unza26-csc4792-chingola_citizen_engagement_communication_plan.csv	5	7
db-unza26-csc4792-chingola_capital_projects_performance_2025_semiannual.csv	13	5
db-unza26-csc4792-chingola_revenue_efficiency_2018_2022.csv	5	5
db-unza26-csc4792-chingola_capital_output_2024.csv	10	9
db-unza26-csc4792-chingola_cdf_2024_applications.csv	58	12
db-unza26-csc4792-chingola_semiannual_payments_2025.csv	8	5
db-unza26-csc4792-chingola_debt_arrears_2025_semiannual.csv	16	8

Dataset Filename	Rows	Columns
db-unza26-csc4792-chingola_idp_strategic_outputs_2024_2033.csv	7	4
db-unza26-csc4792-nchanga_cdf_2025_applications.csv	86	6
db-unza26-csc4792-chingola_economic_classification_2024_2026.csv	7	5
db-unza26-csc4792-chingola_budget_output_performance_2025_semiannual.csv	8	5
db-unza26-csc4792-chingola_citizen_engagement_stakeholder_plan.csv	14	8
db-unza26-csc4792-chingola_budget_programmes_2024_2026.csv	10	5
db-unza26-csc4792-chingola_budget_revenue_2024_2026.csv	177	6
db-unza26-csc4792-chingola_cdf_projects_2024_2026.csv	139	5
db-unza26-csc4792-chingola_semiannual_revenue_2025.csv	17	5
db-unza26-csc4792-chingola_cdf_projects_2025.csv	81	5
db-unza26-csc4792-chingola_cash_flow_projections_2023_2032.csv	100	4
db-unza26-csc4792-chingola_budget_output_economic_classification_2024_2026.csv	123	5
db-unza26-csc4792-chingola_internal_audit_2025_semiannual.csv	8	7

Core Feature Attributes

Standardized column names extracted across the dataset include:

- **Internal Audit Variables:** audit_finding, audit_recommendation, management_action_taken, report_period, source_document, section
- **CDF & Projects Variables:** project_name, constituency, ward, allocated_amount, disbursed_amount, project_status, contractor, application_date
- **Budgets & Revenue Variables:** revenue_source, budget_line_item, approved_budget, actual_revenue, expenditure_amount, financial_year, variance
- **Strategic Planning Variables:** strategic_objective, idp_pillar, target_output, kpi_description, stakeholder_group

Experimental Design, Materials and Methods

1. Data Source & Extraction

All records were sourced directly from public documents published on the official Chingola Municipal Council portal (<https://www.chingolacouncil.gov.zm>). PDF reports, output-based annual budgets, and web tables were parsed into 23 separate raw CSV data files.

2. Preprocessing & Data Cleaning

Data preprocessing was conducted in Google Colab using Python (pandas, re, glob, os, zipfile). The cleaning script executed the following workflow:

1. **Column Standardization:** A regex function `standardise_column_name()` removed special characters, stripped whitespace, converted text to lowercase, and inserted underscores between word boundaries to generate clean `snake_case` headers:

```
def standardise_column_name
    str
        r'([a-z0-9])([A-Z])' r'\1_2'
        r'^a-zA-Z0-9_' ' '
        r'_+' ' '
    return
```

2. **Missing Value Harmonization:** Empty strings, whitespace variations, and string placeholders ("null", "na", "none", "pd.NA") were converted to formal NaN values.
3. **Currency Sanitization & Type Conversion:** Numeric fields stored as strings containing currency symbols or punctuation (e.g., "ZMW 15,000.00", "15,000") were cleaned using regex replacements (`str.replace("ZMW", "")`, `str.replace(", ", "")`) and cast to numeric types using `pd.to_numeric()`.

4. **Deduplication:** Dataframes were evaluated using `df.drop_duplicates()`, with row counts verified before and after to ensure zero loss of critical records.
5. **Output Export:** Cleaned files were saved in pipe-delimited format (`|`) adhering strictly to the naming structure `db-unza26-csc4792-[DESCRIPTION].csv`.

Data Accessibility

- **Kaggle Dataset:** <https://www.kaggle.com/datasets/your-group-kaggle-link>
- **GitHub Codebase:** <https://github.com/Liku-94/csc4792-chingola-council-dataset.git>

Ethics Statement

The dataset contains solely public domain municipal information published by Chingola Municipal Council. No non-public personal data, confidential credentials, or private citizen information were collected.

References

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